

A Survey of Optical Networking for AI Data Center

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Abstract—

I. Abstract

Optical networking has emerged as a pivotal solution for addressing the escalating demands of modern data centers, driven by the need for high bandwidth, low latency, and energy efficiency. Traditional electrical packet switching faces limitations in scalability and power consumption, prompting the exploration of optical networking technologies that offer superior capacity and speed. This survey paper provides a comprehensive overview of the current state of optical networking in AI-driven data centers, focusing on key areas such as optical switching and traffic management, optical interconnects and energy efficiency, hybrid switching architectures, and network management and fault tolerance. It examines the integration of machine learning and cognitive computing into optical network management, highlighting the role of intelligent algorithms in optimizing performance, resource allocation, and resilience. The paper also discusses the challenges and opportunities in designing and implementing optical networks tailored for AI applications, emphasizing the importance of scalability, adaptability, and reliability. By synthesizing the latest research and methodologies, this survey aims to provide a holistic understanding of the evolving landscape of data center infrastructure. The findings underscore the transformative potential of optical networking in enabling next-generation data center networks that are more efficient, scalable, and intelligent.

II. Introduction

Optical networking has become a cornerstone of modern data center infrastructure, driven by the escalating demands for high bandwidth, low latency, and energy efficiency [1]. As data centers continue to expand in scale and complexity, the need for advanced networking solutions that can accommodate the diverse and dynamic traffic patterns of cloud computing, artificial intelligence, and large-scale distributed applications has become increasingly critical [2]. Traditional electrical packet switching, while effective for many use cases, faces inherent limitations in terms of power consumption, scalability, and latency, particularly in high-performance environments [3]. Optical networking, with its inherent advantages in capacity and speed, offers a compelling alternative that can address these challenges [4]. This has led to significant research and development in optical circuit switching, hybrid architectures, and intelligent control mechanisms, all of which are essential for building next-generation data center networks

[1]. This survey paper explores the current state of optical networking in AI-driven data centers, focusing on key areas such as optical switching and traffic management, optical interconnects and energy efficiency, hybrid switching architectures, and network management and fault tolerance [1]. It provides a comprehensive overview of the methodologies, technologies, and challenges involved in the design and implementation of optical networks tailored for AI applications [5]. The paper also delves into the integration of machine learning and cognitive computing into optical network management, highlighting the role of intelligent algorithms in optimizing network performance, resource allocation, and resilience. By examining the interplay between optical networking and AI, this survey aims to provide a holistic understanding of the evolving landscape of data center infrastructure [6]. The content of this survey paper is structured to cover the essential aspects of optical networking for AI data centers, beginning with an exploration of optical switching and traffic management strategies, including dynamic routing, traffic prioritization, and network topology optimization [6]. It then examines the role of optical interconnects in reducing power consumption and enabling scalable infrastructure, followed by a discussion of hybrid switching architectures that combine the strengths of optical and electronic switching [4]. The paper also addresses the challenges of network management and fault tolerance, focusing on robust topology engineering and resource allocation strategies. Finally, it explores the integration of AI and machine learning into optical networking, highlighting the potential of intelligent systems to enhance network performance, adaptability, and reliability [7]. This survey paper contributes to the growing body of knowledge on optical networking for AI data centers by synthesizing the latest research, technologies, and methodologies in the field [6]. It provides a detailed analysis of the current state of the art, identifies key challenges, and outlines potential directions for future research. By consolidating findings from a wide range of sources, the paper offers valuable insights for researchers, practitioners, and policymakers seeking to advance the development of next-generation data center networks [8]. The contributions of this survey are intended to foster further innovation and collaboration in the pursuit of more efficient, scalable, and intelligent optical networking solutions for AI-driven environments [4].

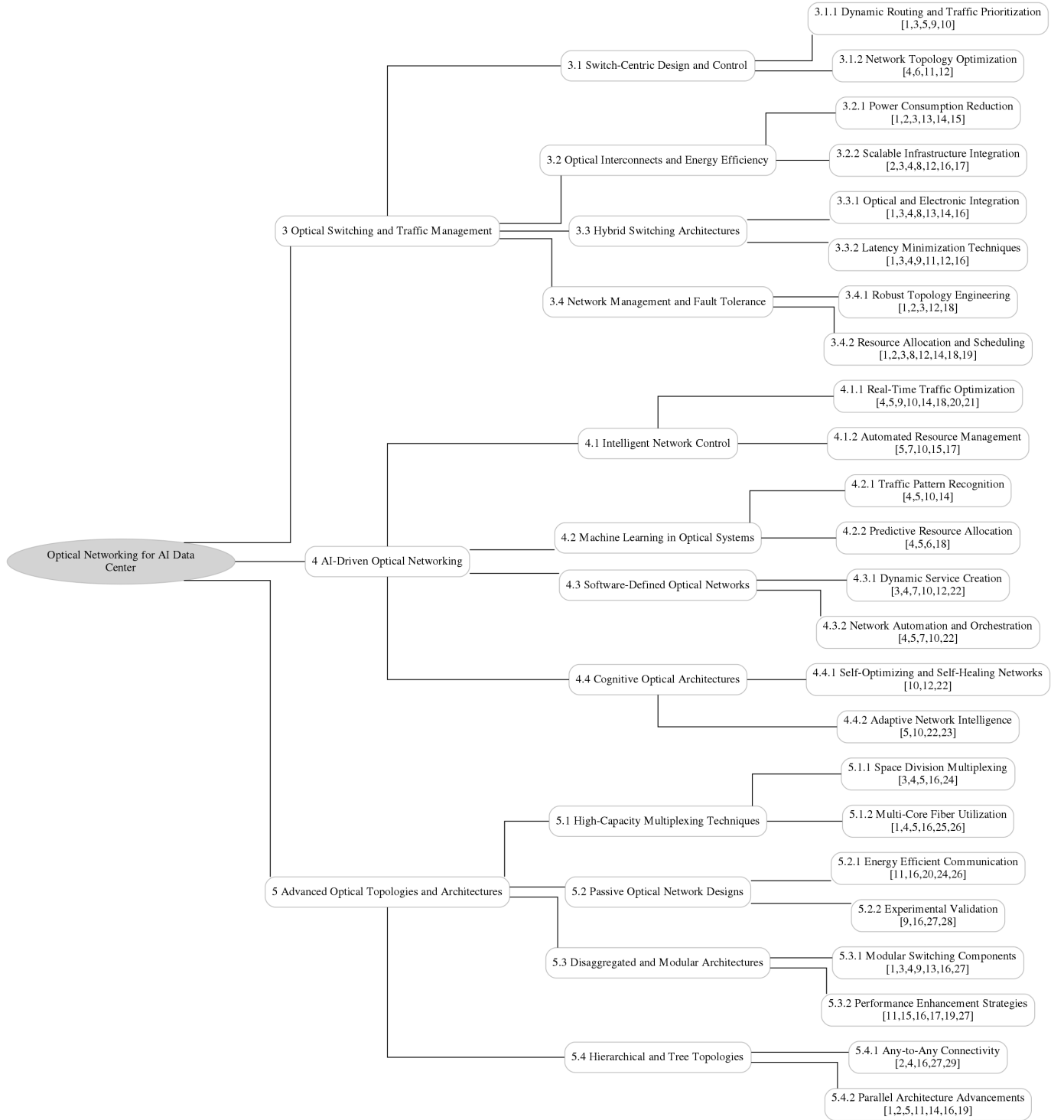


Fig. 1. The outline of our survey: Optical Networking for AI Data Center

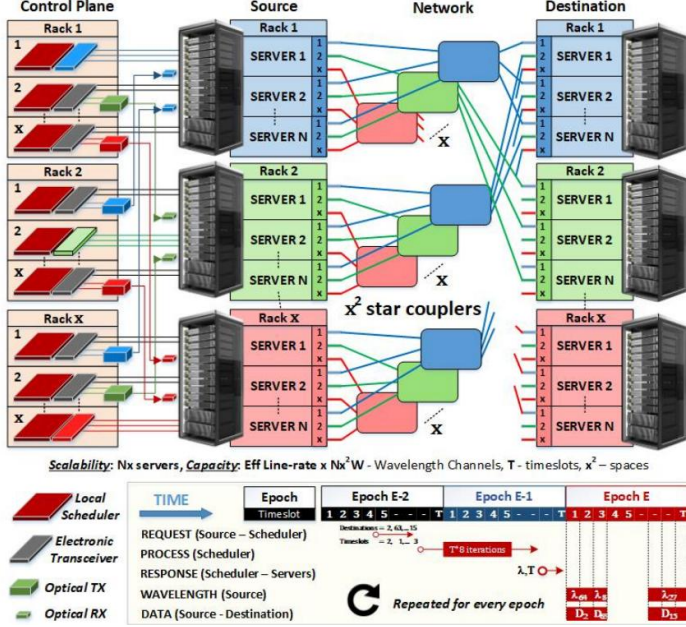


Fig. 2. Architecture of a scalable optical data center network featuring distributed control plane scheduling, star-coupler-based wavelength routing, and time-slotted transmission epochs across server racks.

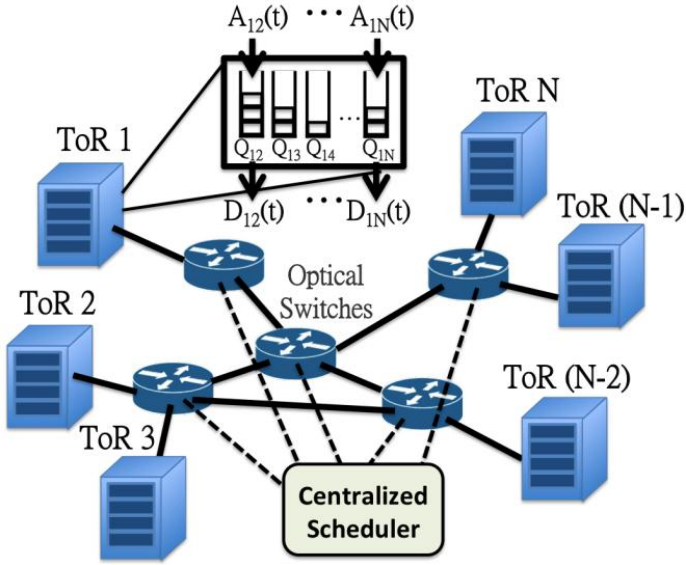


Fig. 3. A centralized optical data center network architecture featuring ToR switches interconnected via optical switches, a centralized scheduler, and virtual output queues modeling inter-rack traffic arrivals and departures.

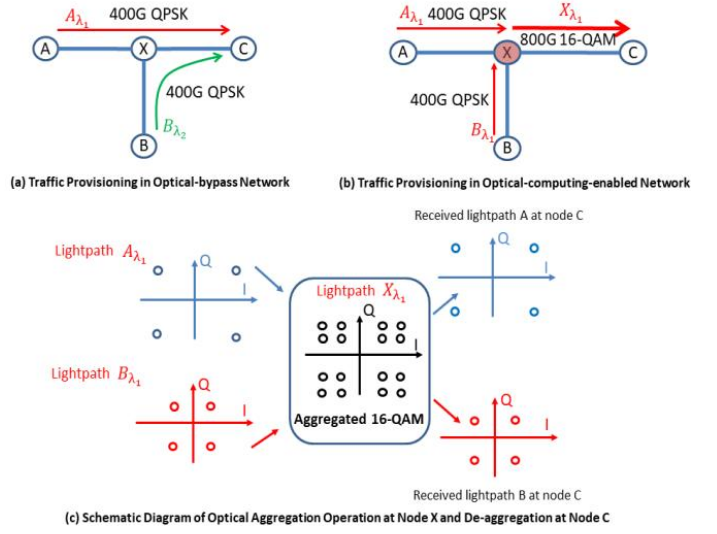


Fig. 4. Comparison of traffic provisioning in optical-bypass versus optical-computing-enabled networks, illustrating the optical aggregation of two 400G QPSK signals into a single 800G 16-QAM signal at intermediate node X and subsequent de-aggregation at the destination.

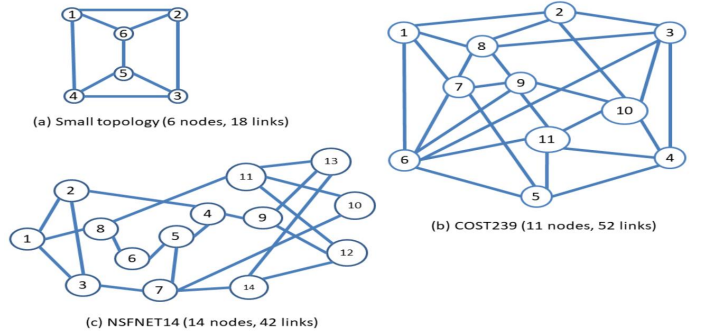


Fig. 5. Three standard network topologies used as benchmarks in network research: (a) Small topology (6 nodes, 18 links), (b) COST239 (11 nodes, 52 links), and (c) NSFNET14 (14 nodes, 42 links).

III. Optical Switching and Traffic Management

A. Switch-Centric Design and Control

1) **Dynamic Routing and Traffic Prioritization:** Dynamic routing and traffic prioritization play a critical role in optimizing the performance of data center networks (DCNs) by efficiently managing the flow of traffic between different pods [9]. In the context of the network architecture described, stable traffic is typically routed through the Optical Circuit Switch (OCS), while bursty traffic is directed through the Electrical Packet Switch (EPS) [3]. This separation ensures that the OCS can maintain dedicated circuits for predictable, long-term traffic patterns, whereas the EPS handles the variability and unpredictability of bursty traffic [3]. A real-time circuit scheduler continuously monitors traffic patterns and re-configures the OCS to ensure that circuits are available for

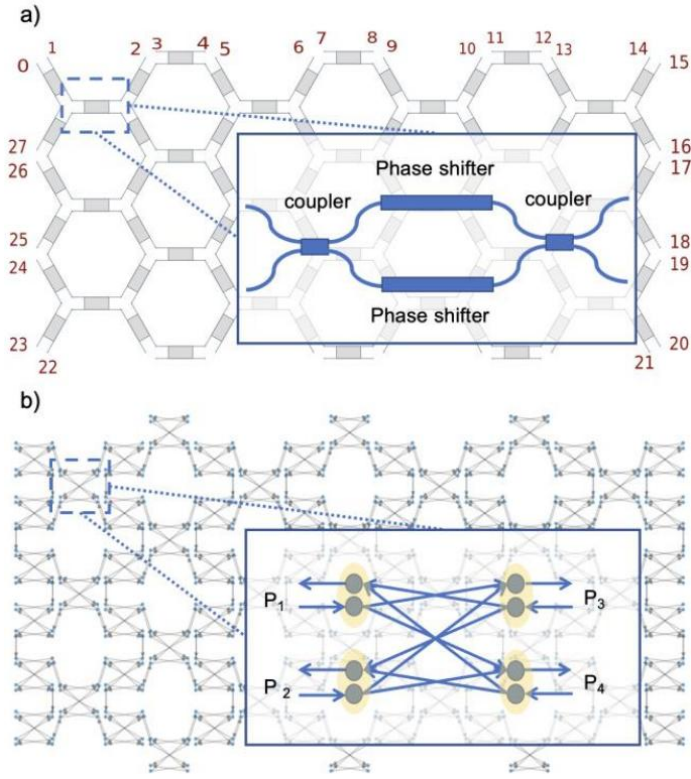


Fig. 6. Schematic diagrams of (a) a hexagonal mesh and (b) a rectangular mesh architecture for programmable photonic integrated circuits based on Mach-Zehnder interferometers.

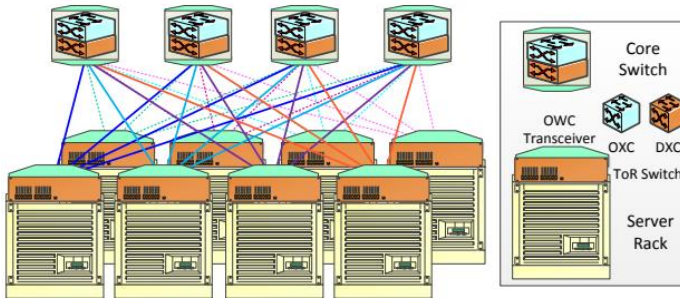


Fig. 7. Architecture of a hybrid optical data center network featuring core switches, OXC/DXC intermediate switches, and ToR-equipped server racks interconnected via optical wireless communication links.

stable traffic, thereby improving overall network reliability and efficiency [3]. To support dynamic routing strategies, a unified workflow for Traffic Assignment (TA) and Traffic Optimization (TO) has been developed. This workflow includes a set of common API functions that allow users to define network topologies and routing strategies through simple Python scripts. These APIs enable both online updates for TA and offline pre-loading for TO, translating high-level network behaviors into low-level OCS configurations. Additionally, an integer linear program (ILP) and a heuristic are employed to minimize system costs while scaling the DCN. The use of an offline routing algorithm further enhances performance by precomputing

paths that minimize latency, leveraging the cyclic nature of circuit connections that are known in advance. The challenges of dynamic routing are compounded by the non-deterministic and load-dependent nature of traffic in DCNs [10]. For instance, Amazon EC2 data centers exhibit significant variations in latency, with a median round-trip time of 600 μ s and a 100th percentile latency of 100 ms [1]. These characteristics necessitate complex methods for addressing, buffer management, and admission control. Furthermore, the increasing volume of intra-DCN traffic, as highlighted by the Cisco Cloud Index, underscores the need for scalable and efficient routing solutions [10]. Dynamic provisioning of lightpaths for stable traffic between server racks using optical networks offers a cost-effective alternative, while the integration of advanced routing algorithms like TROD and tVLB enhances resilience and performance in the face of traffic bursts [5].

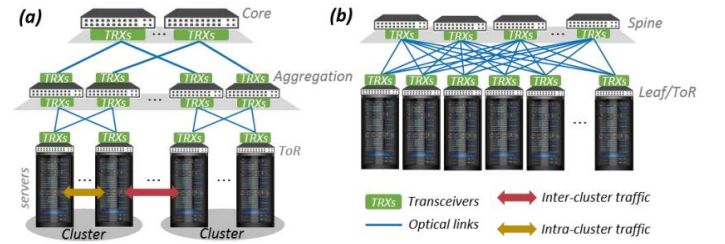


Fig. 8. Comparison of (a) traditional three-tier hierarchical data center network architecture with core, aggregation, and top-of-rack (ToR) layers versus (b) spine-leaf architecture, illustrating optical interconnects, transceiver placements, and intra-/inter-cluster traffic patterns.

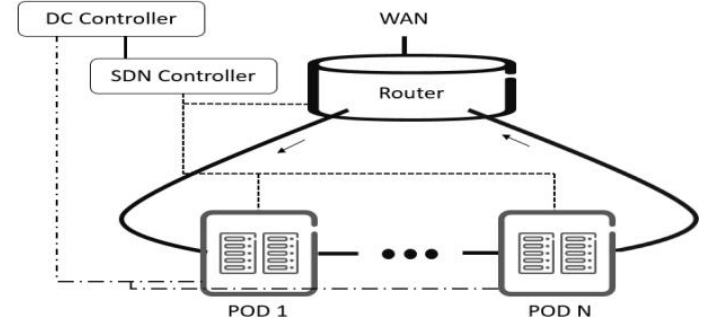


Fig. 9. A software-defined data center (SDN) architecture depicting centralized control plane management where the SDN controller orchestrates network connectivity between the WAN router and distributed server pods (POD 1 to POD N).

2) Network Topology Optimization: Network topology optimization in data center networks is a critical area of research aimed at improving performance, scalability, and cost efficiency [11]. Traditional static topologies, such as fat-tree and torus, often struggle with dynamic traffic patterns and high congestion, necessitating the development of more adaptive and flexible designs. Recent approaches focus on reconfigurable topologies that allow for dynamic adjustments based on traffic demands, leveraging optical

technologies to enable rapid and efficient reconfiguration [4]. These optimizations aim to balance traffic load, reduce latency, and enhance fault tolerance, while also addressing the limitations of conventional electrical switching infrastructures. Key challenges in network topology optimization include the trade-off between hardware complexity and performance gains, as well as the need for efficient routing and resource allocation strategies. Techniques such as component set clustering and traffic-aware partitioning are employed to group nodes with high inter-node communication, thereby minimizing congestion and improving overall throughput. Additionally, the integration of optical switching technologies, such as wavelength and space division multiplexing, enables more flexible and scalable network architectures [4]. This allows for the creation of topologies that can dynamically adapt to changing traffic conditions, ensuring optimal resource utilization and reduced operational costs. The optimization process also involves addressing the impact of topology reconfiguration on network stability and service continuity [12]. Sub-OWD (One-Way Delay) time slices introduce constraints that require careful coordination between routing and topology adjustments to avoid disruptions [12]. Research in this area explores co-design approaches that align routing strategies with topology changes, ensuring minimal latency and high reliability. By integrating advanced algorithms for traffic prediction and resource allocation, network topology optimization continues to evolve, offering promising solutions for next-generation data center networks [6].

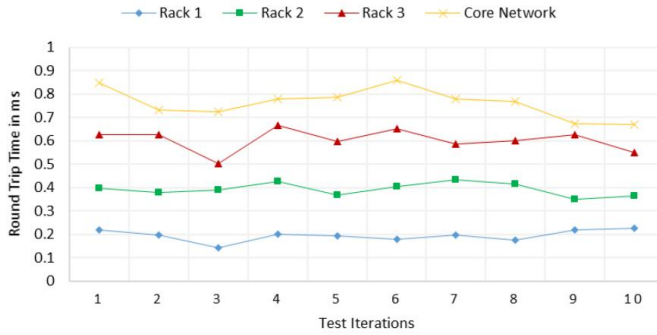


Fig. 10. Round trip time measurements across Rack 1, Rack 2, Rack 3, and Core Network over 10 test iterations.

B. Optical Interconnects and Energy Efficiency

1) **Power Consumption Reduction:** Optical circuit switching (OCS) has emerged as a promising solution for reducing power consumption in data center networks (DCNs) by leveraging the inherent advantages of optical components over traditional electronic switches [1]. The integration of OCS with electrical packet switching, as demonstrated in the Helios prototype, enables significant reductions in both equipment cost and power usage. This is primarily due to the reduced number of switches and

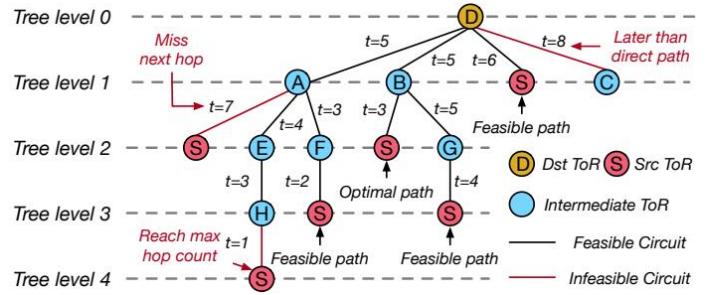


Fig. 11. Circuit routing tree in a multi-tier data center network illustrating optimal paths, feasible circuits, and infeasible routes due to timing constraints or maximum hop count limits.

the lower power requirements of optical interconnects compared to copper-based links [13]. Studies have shown that replacing copper links with optical interconnects can lead to substantial power savings, with some research indicating that over 70

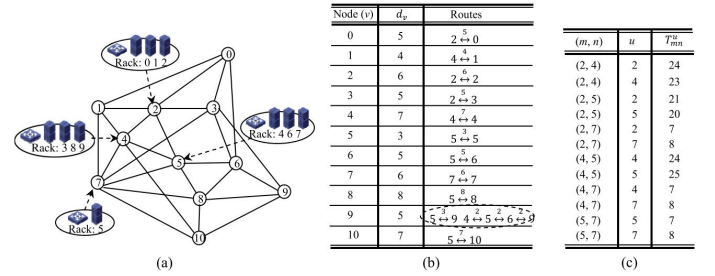


Fig. 12. "A data center network topology with server rack placements and corresponding routing tables showing node traffic demands, routes, and link flow assignments."

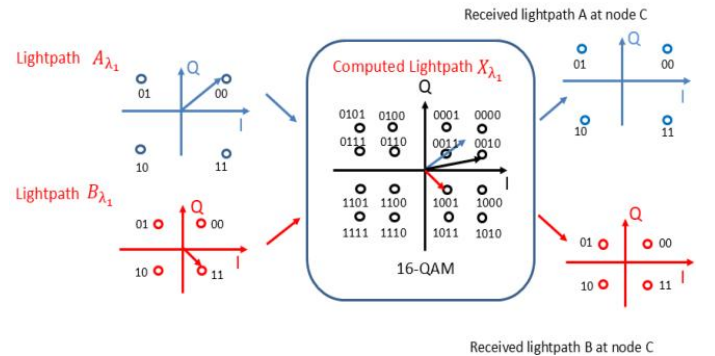


Fig. 13. Physical-layer network coding scheme illustrating the superposition of two QPSK-modulated lightpaths into a 16-QAM constellation and their separation at the receiving node.

2) **Scalable Infrastructure Integration:** Scalable infrastructure integration in modern data center networks (DCNs) is a critical challenge driven by the increasing demand for high bandwidth, low latency, and flexible resource allocation [8]. The Helios prototype, which partitions physical switches into smaller, more manageable

units, exemplifies a strategy to balance network performance and scalability [3]. This approach enables more granular control over traffic flows and facilitates dynamic reconfiguration, which is essential for handling the variable workloads typical in cloud and machine learning environments. By integrating optical circuit switching (OCS) with traditional packet switching, the architecture supports both connection-rich intra-rack and locality-friendly inter-rack communication, addressing the diverse needs of emerging applications [3]. However, the integration of such hybrid architectures introduces complexities in network management and performance optimization [16]. The distributed placement of component sets, while beneficial for load balancing, can lead to increased transmission delay and cost. To mitigate this, a cost scaling factor θ is introduced, translating additional delay into a measurable cost that can be incorporated into routing and scheduling decisions. This factor allows for flexible control over the trade-off between network efficiency and resource utilization, ensuring that the system remains adaptive to changing traffic patterns [17]. The use of θ also highlights the need for intelligent algorithms that can dynamically adjust to network conditions without compromising overall performance. Despite these advancements, the transition to scalable infrastructure remains challenging due to the disruptive nature of reconfigurable optical architectures [4]. These designs often require a complete overhaul of existing network stacks, making deployment a long-term endeavor [12]. Moreover, the reliance on optical interconnects necessitates higher data rates, which are only feasible with optical fibers due to the limitations of electrical transmission [3]. As data centers continue to grow in scale and complexity, the integration of scalable, reconfigurable infrastructure will be essential for supporting the next generation of high-performance computing and cloud services [2].

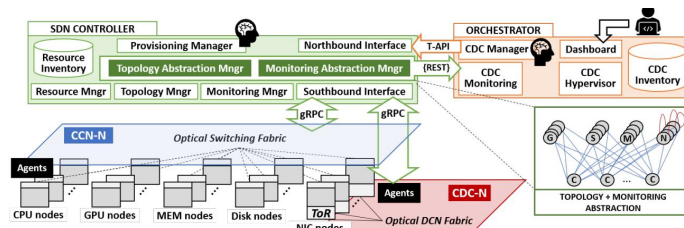


Fig. 14. Architecture of an SDN-orchestrated composable disaggregated infrastructure featuring optical switching fabric, resource abstraction managers, and hierarchical control plane for managing heterogeneous CPU, GPU, memory, disk, and NIC resource nodes.

C. Hybrid Switching Architectures

1) **Optical and Electronic Integration:** Optical and electronic integration in data center networks (DCNs) represents a critical convergence of technologies aimed at addressing the growing demands for bandwidth, energy efficiency, and scalability [8]. This integration leverages

optical circuit switching (OCS) to complement traditional electronic packet switching, enabling dynamic and high-capacity data routing [3]. By combining the strengths of both domains, hybrid architectures reduce power consumption and equipment costs while maintaining the flexibility needed for diverse traffic patterns. The use of optical switches allows for the direct transmission of data in the optical domain, eliminating the need for frequent optical-to-electrical conversions, which enhances performance and reduces latency [3]. This synergy between optical and electronic components is essential for supporting the evolving requirements of next-generation DCNs [14]. The integration of optical and electronic systems introduces unique challenges, particularly in terms of reconfiguration delays and the need for advanced control mechanisms. Optical switches, such as MEMS-based or wavelength-selective switches, often exhibit longer reconfiguration times compared to their electronic counterparts, which can impact the responsiveness of the network [13]. Additionally, the management of optical circuits requires careful coordination with electronic control planes to ensure efficient resource allocation and traffic management [4]. These challenges necessitate the development of robust protocols and architectures that can seamlessly bridge the optical and electronic domains [16]. Research efforts have focused on optimizing the interplay between these technologies to achieve high throughput, low latency, and reliable operation in complex DCN environments [14]. Furthermore, the integration of optical and electronic components enables new capabilities such as topology engineering and dynamic bandwidth provisioning, which are crucial for handling the highly skewed traffic patterns common in data centers [1]. By leveraging optical circuit switching, DCNs can adapt their internal topologies to match traffic demands, improving overall efficiency and resource utilization [14]. This flexibility also supports the deployment of advanced networking protocols and virtualization techniques, enhancing the scalability and manageability of the network. As the field continues to evolve, the successful integration of optical and electronic technologies will play a pivotal role in shaping the future of high-performance, energy-efficient data center networks [1].

2) **Latency Minimization Techniques:** Latency minimization in optical circuit switching (OCS) networks is a critical area of research due to the inherent advantages of optical technologies in reducing transmission delays [3]. OCS systems, being bufferless and eliminating optical-to-electrical (O-E-O) conversions, inherently offer lower latency compared to traditional electrical packet switching [3]. This makes them particularly suitable for applications requiring deterministic and low-latency communication [1]. However, the dynamic nature of data center traffic, especially with the emergence of latency-sensitive "micro flows," necessitates advanced techniques for managing and minimizing latency [9]. These techniques often involve

optimizing the scheduling of optical circuits and ensuring rapid reconfiguration of the network to accommodate changing traffic patterns [4]. To address latency challenges, researchers have explored fast-switched OCS designs that operate on microsecond or sub-microsecond time scales [3]. These designs leverage fine-grained reconfiguration capabilities of optical switches to dynamically allocate resources and reduce the time required to establish connections [16]. Such systems enable more efficient handling of bursty traffic and improve overall network responsiveness. Additionally, latency minimization strategies often involve intelligent scheduling algorithms that account for reconfiguration delays and inter-frame gaps, ensuring that traffic is routed through the most optimal paths [1]. These policies aim to minimize the impact of non-zero reconfiguration times, which can otherwise introduce significant latency in the network. Another key aspect of latency minimization is the coordination of packet buffering and scheduling at endpoints to align with circuit availability [1]. This involves careful planning of traffic flows to ensure that packets are queued and transmitted in a manner that avoids unnecessary delays [12]. Furthermore, the design of optical networks must consider the trade-offs between full bisection bandwidth and resource utilization, as over-provisioning can lead to inefficiencies [4]. Techniques such as component set clustering and wavelength assignment also play a role in optimizing latency by reducing the number of hops and improving the efficiency of data transmission. Overall, the integration of these strategies is essential for achieving the low-latency performance required by modern data center applications [11].

D. Network Management and Fault Tolerance

1) **Robust Topology Engineering:** Robust topology engineering is a critical component in modern data center networks, particularly in environments where dynamic traffic patterns and unpredictable workloads necessitate adaptive infrastructure. This section explores the methodologies and principles that underpin the design of resilient network topologies capable of maintaining performance and reliability under varying conditions. The focus is on architectures that allow for rapid reconfiguration, such as the Helios prototype, which partitions physical switches into smaller, manageable entities to achieve a more balanced and responsive network structure [3]. These designs are essential for supporting high-bandwidth applications and ensuring efficient resource utilization, even as traffic demands fluctuate over time [18]. A key challenge in robust topology engineering lies in the ability to accommodate both predictable and unforeseen traffic patterns. While many systems rely on historical data to anticipate traffic flows and optimize network configurations, the presence of unpredictable bursts can still lead to congestion and performance degradation. To address this, advanced frameworks such as COUDER have been developed, which leverage convex traffic sets to provide

strong throughput guarantees and employ desensitization techniques to manage out-of-bounds traffic. These approaches ensure that network topologies remain effective even when faced with deviations from expected traffic behavior, thereby enhancing overall system resilience [12]. The integration of optical technologies further enhances the robustness of data center topologies by enabling flexible and high-capacity communication paths [1]. Systems like Proteus DCN utilize optical wavelength and space switching to achieve dynamic reconfigurability, offering improved bandwidth and reduced cabling complexity. This capability is crucial for supporting the evolving needs of data centers, where traditional static topologies may struggle to meet the demands of modern applications [2]. By combining optical switching with intelligent topology engineering, future data center networks can achieve greater adaptability, efficiency, and reliability in the face of complex and changing traffic environments [1].

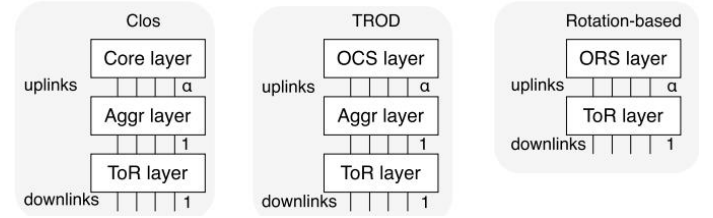


Fig. 15. Comparison of three data center network architectures (Clos, TROD, and Rotation-based) illustrating their hierarchical layer structures and connectivity patterns.

2) **Resource Allocation and Scheduling:** Resource allocation and scheduling in data center networks (DCNs) play a critical role in ensuring efficient utilization of network and computational resources while meeting performance and scalability demands [8]. The dynamic nature of traffic patterns, ranging from stable to bursty, necessitates adaptive mechanisms that can reconfigure network paths and allocate resources in real-time [18]. This involves coordinating packet buffering and scheduling at endpoints to align with circuit availability, particularly in architectures that employ optical circuit switching (OCS) for stable traffic and electronic packet switching (EPS) for bursty traffic [3]. Effective resource allocation strategies must account for both the temporal and spatial distribution of traffic to optimize latency, throughput, and overall system efficiency. The complexity of resource allocation is further compounded by the need to balance network load and reduce congestion, especially in large-scale DCNs where full bisection bandwidth is often unnecessary [14]. This leads to a challenge in provisioning network resources without over-allocating, as certain parts of the network may remain underutilized [12]. Additionally, the integration of optical and electronic switching technologies introduces new dimensions to scheduling, requiring careful coordination between circuit reconfiguration and packet forwarding [1]. Techniques such as topology-routing co-design and

time-slice management are employed to address these challenges, ensuring that network resources are allocated in a manner that supports both high-performance and cost-effective operations. Optimizing resource allocation and scheduling also involves addressing the limitations imposed by hardware scaling and power constraints [19]. As data centers grow in size and complexity, the need for efficient algorithms that can handle large-scale network topologies becomes increasingly important [2]. This includes the development of heuristic and sub-optimal policies that can manage traffic grooming, flow prioritization, and resource clustering. These approaches aim to enhance network utilization while minimizing latency and operational costs, making them essential components in the design of next-generation DCNs [14].

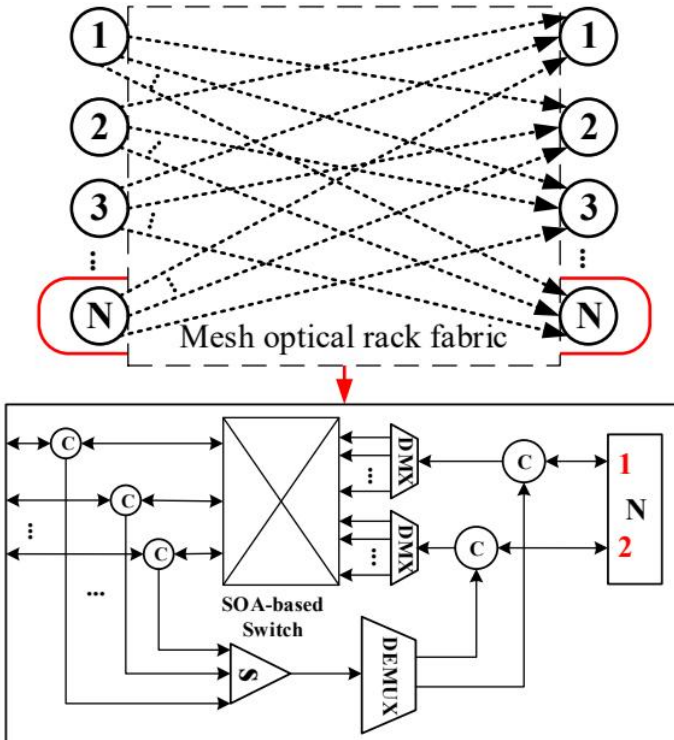


Fig. 16. Architecture of a mesh optical rack fabric interconnect (top) and detailed node design featuring SOA-based switches with wavelength demultiplexers (bottom).

IV. AI-Driven Optical Networking

A. Intelligent Network Control

1) **Real-Time Traffic Optimization:** Real-time traffic optimization in optical packet and circuit switched networks is essential for managing the exponential growth of internet traffic and ensuring efficient resource utilization [5]. This section explores the integration of advanced algorithms and control mechanisms that enable dynamic adjustment of network resources in response to fluctuating traffic demands. By leveraging software-defined networking (SDN) principles, real-time traffic optimization facilitates centralized control and programmability,

allowing for rapid reconfiguration of network paths and bandwidth allocation [10]. This capability is crucial in maintaining high performance and low latency, particularly in data centers and high-speed communication infrastructures where traffic patterns are highly variable and unpredictable [9]. The implementation of real-time traffic optimization involves the use of machine learning (ML) and artificial intelligence (AI) techniques to predict and adapt to traffic patterns [14]. These technologies enable the network to autonomously identify congestion points, reroute traffic, and optimize spectral efficiency. Additionally, the introduction of data analytics and telemetry systems enhances the visibility of network operations, allowing for more informed decision-making. By combining these elements, optical networks can achieve a level of responsiveness that is critical for supporting emerging applications such as cloud computing, multimedia streaming, and large-scale machine learning workloads [4]. This approach not only improves the reliability of the network but also reduces operational costs through more efficient resource management [20]. Furthermore, real-time traffic optimization is closely tied to the development of elastic and programmable network architectures that can scale dynamically with traffic demands [18]. This includes the use of optical aggregation and de-aggregation techniques to manage traffic more effectively across different network layers [21]. The integration of these technologies with AI-driven analytics allows for proactive traffic management, minimizing bottlenecks and ensuring consistent service quality. As network traffic continues to grow, the ability to optimize traffic in real time will become increasingly vital for maintaining the performance and scalability of optical networks [5]. This section highlights the key strategies and technologies that underpin real-time traffic optimization, emphasizing their role in shaping the future of high-capacity, flexible, and energy-efficient optical communication systems [4].

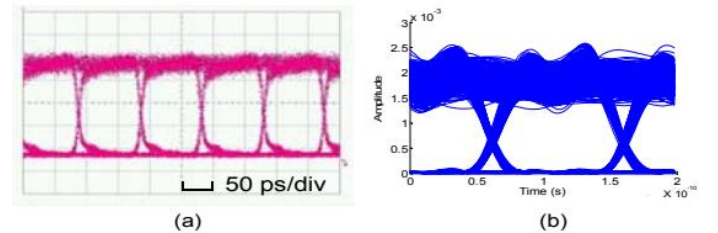


Fig. 17. Eye diagrams illustrating signal quality of a high-speed digital transmission system: (a) measured experimental result and (b) simulation output.

2) **Automated Resource Management:** Automated resource management has emerged as a critical component in modern networked systems, particularly in the context of data centers and optical networks [7]. It involves the dynamic allocation, monitoring, and optimization of computational, storage, and network resources to meet varying workload demands efficiently [17]. This approach

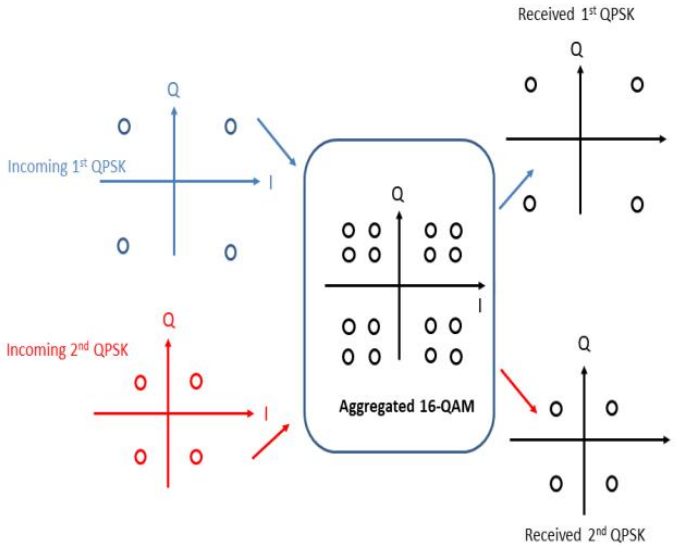


Fig. 18. Illustration of constellation aggregation where two independent QPSK signals are superimposed to form a 16-QAM constellation and subsequently separated at the receiver.

is essential in environments where resource utilization must be maximized while minimizing waste and operational costs. By leveraging advanced algorithms and control mechanisms, automated systems can adapt in real-time to changing conditions, ensuring that resources are allocated where and when they are needed most. This capability is especially important in large-scale distributed systems where manual intervention is impractical and inefficient. The integration of artificial intelligence and machine learning techniques has further enhanced the capabilities of automated resource management. These technologies enable systems to predict future resource needs based on historical data and current trends, allowing for proactive rather than reactive management. Additionally, automated systems can dynamically adjust resource configurations in response to performance metrics, such as latency, throughput, and error rates [17]. This not only improves system efficiency but also enhances reliability and user experience. The use of intelligent decision-making frameworks allows for the optimization of complex resource interactions, ensuring that network and computing resources are utilized in a manner that aligns with both technical and business objectives. In the context of optical networks and data center networks, automated resource management plays a pivotal role in addressing scalability and performance challenges [5]. With the increasing demand for high-speed data transmission and low-latency services, traditional static resource allocation methods are no longer sufficient [15]. Automated systems enable the efficient orchestration of optical and electrical resources, supporting dynamic service provisioning and traffic management [7]. Furthermore, they facilitate the implementation of software-defined networking and net-

work function virtualization, which are key enablers for flexible and agile network architectures [10]. By continuously adapting to evolving traffic patterns and service requirements, automated resource management ensures that network infrastructures remain resilient, efficient, and capable of supporting future technological advancements [5].

B. Machine Learning in Optical Systems

1) Traffic Pattern Recognition: Traffic pattern recognition plays a pivotal role in the design and operation of advanced optical networks, particularly in the context of optical packet and circuit switching [4]. As network traffic continues to grow exponentially, the ability to identify and classify traffic patterns becomes essential for optimizing resource allocation, improving network efficiency, and ensuring quality of service. This section explores the methodologies and techniques used for traffic pattern recognition, focusing on how they enable intelligent decision-making in dynamic network environments. By leveraging statistical analysis, machine learning, and data-driven approaches, network operators can gain insights into traffic behavior, detect anomalies, and predict future trends, which are critical for maintaining network performance and reliability. In optical networks, traffic pattern recognition is often integrated with machine learning algorithms to enhance the accuracy and adaptability of network management systems [14]. These algorithms analyze historical and real-time data to detect recurring traffic patterns, such as those generated by multimedia applications, cloud computing, and parallel processing tasks. The recognition of such patterns allows for the dynamic reconfiguration of network resources, enabling more efficient routing, load balancing, and bandwidth allocation. Furthermore, the integration of traffic pattern recognition with optical computing and intelligent optical nodes facilitates real-time processing of data flows, reducing latency and improving overall network throughput [14]. This synergy between traffic analysis and optical technologies is crucial for supporting the evolving demands of modern communication networks [5]. The application of traffic pattern recognition in optical networks also extends to the optimization of energy consumption and the reduction of operational costs [4]. By identifying traffic hotspots and predicting traffic fluctuations, network operators can implement energy-efficient strategies, such as dynamic power management and resource scaling. Additionally, the use of traffic pattern recognition in conjunction with software-defined networking and network function virtualization enables the creation of flexible and scalable network architectures [10]. These advancements contribute to the development of smart optical networks that are capable of adapting to changing traffic conditions while maintaining high performance and reliability [4]. As the field continues to evolve, further research into traffic pattern recognition

techniques will be essential for addressing the challenges of future optical network infrastructures [14].

2) **Predictive Resource Allocation:** Predictive resource allocation has emerged as a critical enabler for enhancing the efficiency and adaptability of future optical and data center networks [6]. By leveraging advanced analytics and machine learning techniques, this approach aims to forecast traffic patterns and allocate network resources proactively, thereby minimizing latency and maximizing throughput [6]. The integration of AI-driven predictive models allows for dynamic adjustments to network configurations, ensuring optimal utilization of bandwidth and computational resources. This capability is particularly vital in environments where traffic demands fluctuate rapidly, such as in cloud computing and large-scale AI applications. The ability to anticipate and respond to traffic variations in real-time significantly improves the overall performance and reliability of the network infrastructure [18]. The implementation of predictive resource allocation involves complex decision-making processes that balance multiple objectives, including minimizing resource wastage, ensuring quality of service, and maintaining network stability. Techniques such as machine learning, statistical modeling, and optimization algorithms are commonly employed to build accurate predictive models. These models are trained on historical traffic data and network performance metrics to identify patterns and trends that inform future resource allocation strategies. Additionally, the use of reinforcement learning and deep learning has shown promise in adapting to evolving network conditions and improving long-term resource planning. The challenge lies in achieving a balance between model complexity and computational efficiency, ensuring that predictive models can be deployed in real-time environments without introducing significant overhead. In the context of next-generation optical networks, predictive resource allocation plays a pivotal role in addressing the growing demand for high-capacity, low-latency communication [5]. By integrating predictive capabilities into network management systems, operators can optimize the allocation of optical wavelengths, switching elements, and other critical resources [4]. This not only enhances the efficiency of network operations but also supports the deployment of advanced services such as real-time analytics, edge computing, and distributed AI workloads. The continuous refinement of predictive models through feedback loops and adaptive learning ensures that resource allocation strategies remain effective in the face of changing network dynamics and user demands.

C. Software-Defined Optical Networks

1) **Dynamic Service Creation:** Dynamic service creation refers to the ability of a network to autonomously configure, deploy, and manage services in response to changing user demands and network conditions. This capability is essential in modern optical networks, where

the integration of artificial intelligence (AI) and software-defined networking (SDN) enables real-time adaptation and optimization [4]. By leveraging programmable automation, dynamic service creation allows for the rapid provisioning of virtual network functions (VNFs) and the orchestration of complex service chains, ensuring efficient resource utilization and improved service quality [10]. This process is often guided by intelligent algorithms that analyze traffic patterns, predict future demands, and adjust network configurations accordingly, leading to more agile and responsive network operations. The implementation of dynamic service creation relies on advanced network management frameworks that support flexible resource allocation and service orchestration [10]. These frameworks incorporate elements such as network function virtualization (NFV), SDN controllers, and AI-driven decision-making modules to enable seamless service provisioning [10]. In optical networks, this involves the dynamic allocation of bandwidth, routing of traffic, and management of optical switches to ensure optimal performance [3]. Additionally, the use of machine learning techniques allows for the continuous refinement of service creation strategies, adapting to evolving network topologies and user requirements. This level of automation not only enhances the scalability of optical networks but also reduces operational complexity and costs [7]. Challenges in dynamic service creation include ensuring service reliability, maintaining security, and managing the increased complexity introduced by automated decision-making. Furthermore, the integration of AI into service creation processes requires robust data management and model training to ensure accurate and timely service provisioning. Despite these challenges, the ongoing development of intelligent network architectures and the maturation of AI technologies are paving the way for more efficient and adaptive service creation mechanisms [22]. These advancements are critical for supporting the growing demands of next-generation networks, where flexibility and responsiveness are paramount [12].

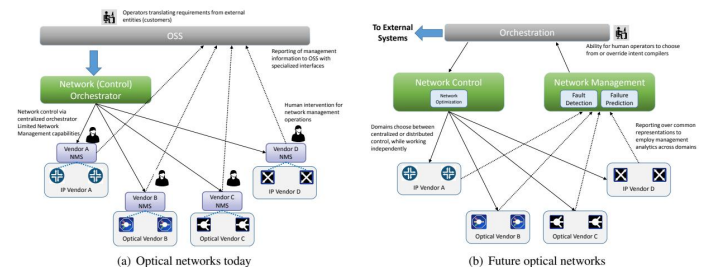


Fig. 19. Architecture comparison illustrating the transition from vendor-specific network management systems in current optical networks to unified orchestration with decoupled control and management planes in future optical networks.

2) **Network Automation and Orchestration:** Network automation and orchestration play a pivotal role in the evolution of modern optical networks, enabling efficient

management, dynamic resource allocation, and rapid service provisioning [7]. As optical networks grow in complexity, the integration of automated control mechanisms becomes essential to handle the increasing demands for scalability, reliability, and adaptability [4]. Automation is achieved through the deployment of intelligent software-defined networking (SDN) frameworks and network functions virtualization (NFV), which allow for centralized control and policy-driven management [10]. These technologies facilitate the seamless coordination of optical switching, routing, and resource allocation, ensuring optimal utilization of network infrastructure while minimizing manual intervention [4]. The transition from manual to automated operations is critical in supporting the dynamic and heterogeneous traffic patterns prevalent in next-generation networks. Orchestration extends automation by providing a higher-level abstraction for managing complex network services and ensuring end-to-end service delivery. It involves the coordination of multiple network functions, including optical path provisioning, traffic engineering, and fault management, through a unified control plane [4]. This is often realized using service orchestration platforms that leverage application programming interfaces (APIs) and standardized models to enable interoperability across diverse network elements. The orchestration layer also supports the deployment of advanced functionalities such as network slicing, where distinct virtual networks can be instantiated to meet specific service requirements [10]. By integrating automation and orchestration, optical networks can achieve greater agility, reduce operational costs, and enhance the overall user experience [7]. The integration of artificial intelligence (AI) and machine learning (ML) further enhances network automation and orchestration by enabling predictive analytics, self-optimization, and anomaly detection [22]. These capabilities allow networks to adapt in real-time to changing conditions, such as traffic fluctuations and failures, without human intervention. Additionally, AI-driven automation supports the efficient management of large-scale optical infrastructures, ensuring that resources are allocated in a manner that maximizes performance and minimizes latency [5]. As optical networks continue to evolve, the synergy between automation, orchestration, and AI will be instrumental in shaping the next generation of intelligent and self-managing communication systems [7].

D. Cognitive Optical Architectures

1) Self-Optimizing and Self-Healing Networks: Self-optimizing and self-healing networks represent a critical advancement in the evolution of future communication systems, enabling autonomous adaptation and recovery from failures without human intervention [22]. These networks leverage intelligent algorithms and distributed control mechanisms to dynamically adjust their configurations, optimize resource allocation, and restore services

in response to changing conditions [10]. The integration of machine learning and artificial intelligence has further enhanced the ability of these networks to predict and mitigate potential issues, ensuring continuous and reliable service delivery. By embedding self-awareness and adaptive capabilities, self-optimizing networks can improve efficiency, reduce operational costs, and enhance scalability in complex and dynamic environments. Self-healing capabilities are particularly vital in maintaining network resilience against failures, such as link disruptions, node malfunctions, or traffic congestion [22]. These mechanisms employ proactive monitoring, fault detection, and automated recovery strategies to restore network functionality swiftly. Techniques such as redundancy management, dynamic rerouting, and predictive maintenance are commonly used to ensure minimal service disruption. Additionally, self-optimizing networks utilize real-time data analysis and feedback loops to refine their performance, adjusting parameters like routing paths, bandwidth allocation, and power consumption based on current demand and network state. This level of autonomy significantly reduces the dependency on manual intervention and enhances the overall robustness of the network infrastructure. The development of self-optimizing and self-healing networks is closely tied to the broader goals of future networks, including enhanced scalability, energy efficiency, and service quality [22]. These networks are designed to operate in both centralized and distributed control environments, supporting a wide range of applications from high-speed data transmission to low-latency services. The implementation of such capabilities requires advanced network management frameworks, standardized protocols, and robust security measures to ensure seamless operation. As research in this area continues to evolve, the integration of these autonomous features is expected to play a pivotal role in shaping the next generation of communication networks, making them more intelligent, adaptive, and resilient [12].

2) Adaptive Network Intelligence: Adaptive Network Intelligence (ANI) represents a pivotal advancement in the evolution of future networks, particularly within the context of the new-generation network (NWGN) and its broader vision of Future Networks. ANI integrates self-configuration, self-healing, and self-optimization capabilities, enabling networks to dynamically adapt to changing conditions and operational demands [22]. This paradigm shift is driven by the need for highly resilient and efficient network infrastructures capable of supporting the exponential growth in data traffic and the complexity of emerging applications [5]. By leveraging advanced algorithms and real-time data processing, ANI ensures that network resources are utilized optimally, enhancing both performance and reliability. The implementation of ANI is closely tied to the integration of artificial intelligence (AI) and machine learning (ML) techniques within network management frameworks. These technolo-

gies empower networks to predict and respond to traffic patterns, resource constraints, and potential failures with minimal human intervention [23]. The use of AI in network intelligence facilitates the automation of critical functions such as routing, resource allocation, and fault detection, thereby reducing operational overhead and improving service delivery [22]. Furthermore, ANI supports the dynamic orchestration of virtual network functions (VNFs) and service chains, enabling flexible and scalable network services that align with the demands of modern cloud and edge computing environments [10]. Key challenges in the development of ANI include ensuring interoperability across heterogeneous network components, maintaining security and privacy in automated decision-making processes, and managing the computational complexity of AI-driven algorithms. Addressing these challenges requires the design of robust, standardized, and secure frameworks that can support the seamless integration of ANI into existing and future network architectures. As research in this area continues to advance, the potential of ANI to transform network operations and service delivery remains a central focus in the pursuit of next-generation communication systems.

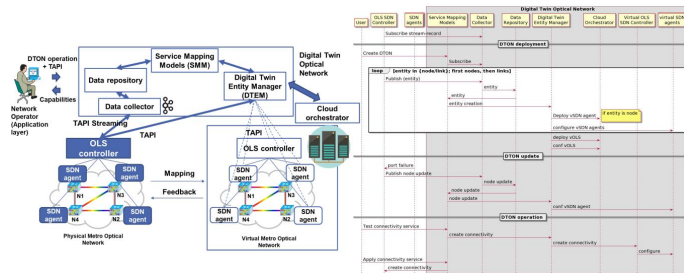


Fig. 20. System architecture and operational workflows for a Digital Twin Optical Network (DTON) framework, illustrating the mapping between physical and virtual metro optical networks along with deployment, update, and operation sequences.

V. Advanced Optical Topologies and Architectures

A. High-Capacity Multiplexing Techniques

1) **Space Division Multiplexing:** Space Division Multiplexing (SDM) has emerged as a critical technique for enhancing the capacity and efficiency of optical communication systems. By enabling the simultaneous transmission of multiple data streams through distinct spatial channels within a single fiber, SDM offers a promising solution to the growing demand for high-bandwidth connectivity [5]. This approach leverages advanced fiber architectures such as few-mode fibers (FMF) and multicore fibers (MCF), allowing for the independent routing of signals across different spatial modes or cores [24]. The implementation of SDM requires sophisticated signal processing techniques to mitigate challenges like modal interference and differential mode dispersion, which are inherent in multi-mode transmission [24]. These technical complexities underscore the

need for continued research into robust and scalable SDM solutions that can support the evolving requirements of modern communication networks. The integration of SDM into optical circuit switching (OCS) topologies has been explored as a means to improve resource utilization and reduce latency in data center architectures [3]. By enabling parallel data transmission across multiple spatial channels, SDM facilitates more efficient use of fiber infrastructure, thereby supporting the modular and scalable design of composable data centers [5]. This is particularly relevant in the context of the Box-modular topology, where independent modules are interconnected through optical backplanes [16]. The use of SDM in such configurations allows for enhanced flexibility in resource allocation and improved network performance, making it a key enabler for next-generation optical switching architectures [4]. Furthermore, SDM complements other multiplexing techniques such as wavelength division multiplexing (WDM) and time division multiplexing (TDM), offering a multi-dimensional approach to capacity expansion. Despite its potential, the deployment of SDM faces several technical and practical challenges. These include the need for precise alignment and coupling of spatial channels, the complexity of digital signal processing (DSP) for mode separation, and the cost associated with advanced fiber and transceiver technologies [24]. Additionally, the integration of SDM into existing network infrastructures requires careful planning and optimization to ensure compatibility and reliability [16]. Ongoing research aims to address these challenges through innovations in fiber design, signal processing algorithms, and network management strategies [4]. As the demand for higher capacity and lower latency continues to grow, SDM is expected to play an increasingly vital role in shaping the future of optical communication systems [5].

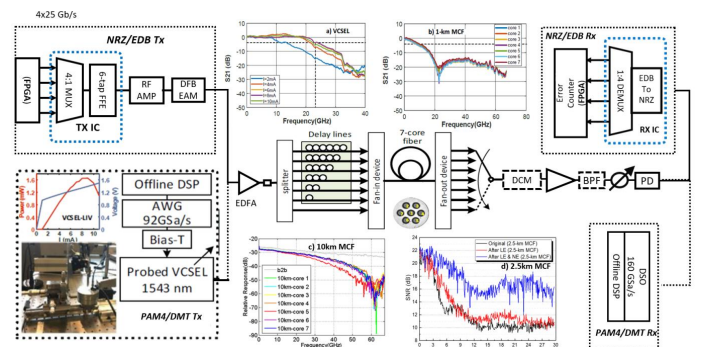


Fig. 21. Experimental setup and frequency response characterization for high-speed optical transmission over 7-core multi-core fiber using NRZ/EDB and PAM4/DMT modulation formats with corresponding transmitter and receiver architectures.

2) **Multi-Core Fiber Utilization:** Multi-core fiber utilization represents a critical advancement in addressing the escalating demands of data center networks, particularly in enhancing capacity and reducing cabling complexity [5]. By leveraging spatial division multiplexing (SDM),

multi-core fibers enable parallel data transmission across multiple cores within a single fiber, significantly boosting the overall bandwidth [5]. This approach not only increases the data rate per fiber but also minimizes the number of physical connections required, thereby simplifying infrastructure and improving scalability [25]. The integration of multi-core fibers into disaggregated data center architectures facilitates more efficient resource allocation and dynamic reconfiguration, aligning with the need for flexible and high-performance computing environments [26]. The deployment of multi-core fibers in optical interconnects plays a pivotal role in supporting the high-speed, low-latency requirements of modern data centers [16]. These fibers are particularly advantageous in long-haul and short-reach applications, where high capacity and reliability are essential. By enabling higher data densities and reducing signal degradation, multi-core fibers contribute to improved network performance and energy efficiency [5]. Additionally, their compatibility with existing optical communication technologies allows for seamless integration into current infrastructure, making them a viable solution for future-proofing data center networks against growing traffic demands [1]. Furthermore, the utilization of multi-core fibers is closely tied to the development of advanced optical networking technologies, such as passive optical networks (PON) and wavelength division multiplexing (WDM) [4]. These technologies, when combined with multi-core fibers, offer enhanced flexibility and resource optimization, supporting the complex and diverse requirements of data center applications. As research continues to refine the design and deployment of multi-core fibers, their potential to revolutionize data center architectures and optical communication systems becomes increasingly apparent, driving the evolution of next-generation network infrastructures [4].

B. Passive Optical Network Designs

1) **Energy Efficient Communication:** Energy efficient communication in disaggregated data center architectures presents a critical challenge due to the increased complexity and distributed nature of the system [11]. These architectures require lower latencies compared to traditional direct-attached systems, necessitating optimized communication protocols and hardware designs. The need for higher bandwidth and bandwidth density further complicates the energy efficiency landscape, as increased data rates often lead to higher power consumption [11]. To address these issues, research has focused on developing communication strategies that minimize energy use while maintaining performance, including the use of low-power transceivers, efficient routing algorithms, and dynamic resource allocation techniques [20]. The design of energy-efficient communication systems must also consider the physical infrastructure and the orchestration of distributed hardware modules [16]. While much of the existing work has focused on the implementation and performance eval-

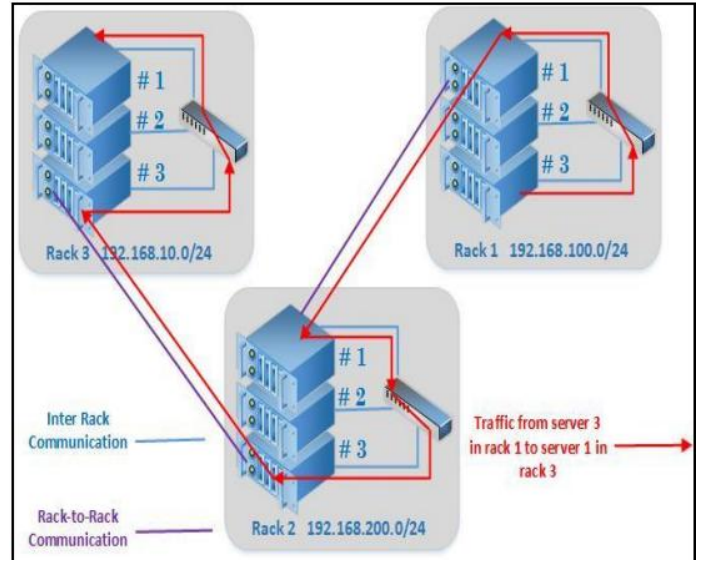


Fig. 22. Multi-rack data center network topology illustrating inter-rack and rack-to-rack communication paths across three server racks with distinct IP subnets (192.168.10.0/24, 192.168.100.0/24, and 192.168.200.0/24).

uation of physical infrastructure, there is a growing need to address the unique challenges introduced by disaggregation. These include managing the energy consumption of interconnects, optimizing power usage across distributed components, and ensuring that communication protocols are adapted to the dynamic and heterogeneous nature of disaggregated systems. Effective energy management in such environments requires a holistic approach that integrates hardware, software, and network-level optimizations. Furthermore, the evolution of communication technologies, such as optical wireless communication (OWC) and advanced modulation formats, offers promising avenues for achieving energy efficiency. OWC systems, for example, leverage abundant bandwidth and low-cost components to support high data rates with reduced power consumption [11]. Similarly, the use of advanced modulation schemes like PAM4 and DMT enables high-speed transmission with lower system complexity [24]. These innovations, when integrated into disaggregated architectures, can significantly enhance energy efficiency while meeting the demands of next-generation data centers and communication networks [26].

2) **Experimental Validation:** Experimental validation plays a critical role in assessing the feasibility and effectiveness of distributed data center (DDC) architectures, particularly in demonstrating their performance improvements over traditional server-centric models [9]. This section focuses on the empirical evaluation of DDC systems, where key metrics such as latency, throughput, and energy efficiency are measured under controlled conditions. The results consistently show a 10-fold increase in performance compared to conventional architectures,

alongside a significant improvement in performance-per-dollar ratio, up to 87

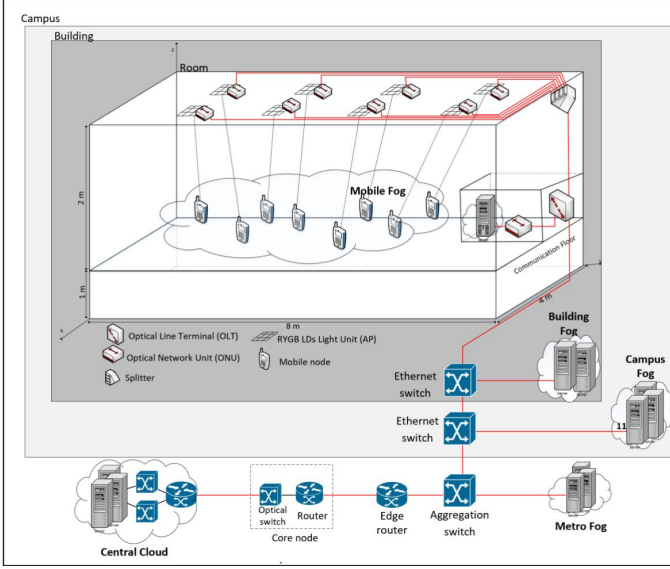


Fig. 23. A hierarchical fog computing architecture integrating visible light communication (LiFi) access points with passive optical network (PON) infrastructure to support mobile edge computing across mobile, building, campus, and metro tiers in a smart campus environment.

C. Disaggregated and Modular Architectures

1) **Modular Switching Components:** Modular switching components play a critical role in enabling the flexibility and scalability of disaggregated data center architectures [27]. These components must support high bandwidth, low latency, and efficient resource allocation across distributed hardware modules [16]. Traditional switching architectures often struggle to meet the stringent requirements imposed by such systems, necessitating the development of specialized modular switches that can dynamically adapt to varying workloads [9]. These switches are typically designed with a focus on minimizing latency while maintaining high throughput, ensuring that the communication between modules remains efficient and reliable [13]. The integration of optical technologies further enhances the capabilities of these switching components by providing the necessary bandwidth and scalability for modern data center applications [1]. The design of modular switching components involves addressing several technical challenges, including the need for cost-effective solutions that reduce power consumption without compromising performance. This is particularly important in large-scale deployments where energy efficiency and operational costs are significant concerns. Additionally, the switching infrastructure must support advanced features such as traffic prioritization, virtualization, and dynamic reconfiguration to accommodate diverse application requirements. These capabilities are essential for enabling the orchestration of modular

resources, allowing for the creation of customized computational nodes that can be optimized for specific workloads [16]. As a result, the development of modular switching components requires a balance between hardware innovation and software-driven management strategies. Optical communication technologies are increasingly being leveraged to enhance the performance of modular switching components, offering higher bandwidth and lower latency compared to traditional electrical interconnects [3]. However, the adoption of optical solutions introduces new challenges, such as the need for specialized optical components and the complexity of integrating them into existing infrastructures [4]. Despite these challenges, the potential benefits of optical-based modular switching, including improved scalability and energy efficiency, make them a promising direction for future research and development [3]. The continued refinement of these components will be crucial in realizing the full potential of disaggregated and composable data center architectures.

2) **Performance Enhancement Strategies:** Performance enhancement strategies in distributed data center (DDC) architectures have gained significant attention due to the need for improved scalability, efficiency, and cost-effectiveness [16]. A 10-fold increase in performance over traditional server-centric designs has been reported, alongside an improvement in performance-per-dollar by up to 8.7

D. Hierarchical and Tree Topologies

1) Any-to-Any Connectivity:

$f_i(x)$ is the local objective function of node i .

(1)

The 3-tier tree topology incorporating dBOSM, dROSM, and dDOSM switches is designed to enable any-to-any connectivity by structuring the network in a hierarchical manner that supports scalable and efficient data routing [27]. At the lowest tier, dBOSM switches manage local connectivity within individual server boxes, while dROSM switches aggregate traffic across multiple boxes within a rack [27]. At the highest tier, dDOSM switches interconnect racks and provide the backbone for inter-rack communication, ensuring that any node can establish a direct path to any other node [27]. This layered architecture minimizes the number of hops and reduces latency, while also enabling dynamic reconfiguration to accommodate varying traffic patterns. The use of optical interconnects further enhances the performance by providing high bandwidth and low signal degradation over long distances [4]. The implementation of this topology requires careful orchestration of both hardware and software components to maintain seamless connectivity across the network [16]. The disaggregation of optical switch modules introduces additional complexity in terms of configuration and management, as each switch must be properly aligned and synchronized to ensure end-to-end communication. This involves not only setting up

intra-node connections but also establishing inter-node links that can dynamically adapt to traffic demands. The control plane must therefore support fine-grained resource allocation and fault tolerance mechanisms to maintain reliability and performance under varying load conditions. These challenges necessitate advanced management frameworks that can efficiently monitor and manage the distributed components of the network [29]. The benefits of this approach are significant, particularly in large-scale data center environments where high throughput and low latency are critical [2]. By enabling direct communication between any two nodes, the 3-tier tree topology reduces the dependency on centralized switching elements and distributes the load more evenly across the network. This leads to improved scalability, reduced congestion, and enhanced fault resilience. Additionally, the modular nature of the architecture allows for easier upgrades and maintenance, as individual components can be replaced or upgraded without disrupting the entire system. Overall, this topology represents a compelling solution for achieving flexible and high-performance any-to-any connectivity in next-generation optical networks [4].

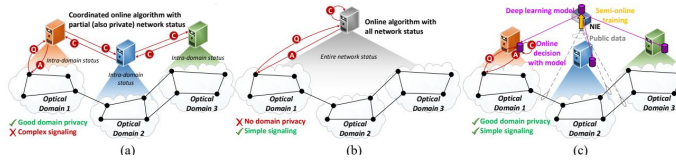


Fig. 24. Comparison of inter-domain routing architectures in multi-domain optical networks, illustrating the trade-offs between domain privacy and signaling complexity across (a) coordinated online algorithms with partial private status, (b) centralized online algorithms with full network status, and (c) deep learning-based online decision models with semi-online training.

2) Parallel Architecture Advancements: Parallel architecture advancements have significantly transformed the design and performance of modern data centers, enabling more efficient resource utilization and scalability [11]. The integration of high-capacity multi-core processors with multithreading capabilities has allowed for increased computational power while maintaining lower power consumption and reduced physical footprints [19]. This shift not only enhances the performance of individual nodes but also supports the deployment of modular and reconfigurable systems, where independent hardware modules can be combined to form tailored computational nodes [16]. Such flexibility is crucial for meeting the diverse and evolving demands of contemporary applications, thereby improving infrastructure utilization and system modularity. Optical technologies have emerged as a critical enabler for achieving the stringent connectivity requirements between distributed hardware modules in parallel architectures [16]. By leveraging optical interconnections, data centers can support high-bandwidth, low-latency communication across extended distances, which is essential for maintaining performance in large-scale

distributed systems [1]. The use of optical technologies also aligns with the need for cost-effective, energy-efficient, and scalable solutions, particularly when combined with advanced parallel processing techniques [16]. This synergy between optical connectivity and parallel computing architectures paves the way for more robust and adaptable data center infrastructures [14]. Furthermore, the evolution of parallel architectures has been driven by the need to address challenges related to power consumption, cost, and scalability. The adoption of spatial division multiplexing (SDM) techniques has enabled the seamless extension of parallelism over long-term operations, offering significant improvements in data throughput and system capacity [5]. These advancements are essential for supporting the growing demands of a hyper-connected world, where the ability to scale efficiently and maintain high performance is paramount. As data centers continue to evolve, the integration of these parallel architecture innovations will play a vital role in shaping the future of distributed computing [2].

VI. Future Directions

Despite significant advancements in optical networking for AI-driven data centers, several limitations and gaps remain that hinder the full realization of high-performance, scalable, and energy-efficient optical infrastructures. One major limitation is the current inability to dynamically and efficiently manage traffic in heterogeneous environments, where both stable and bursty traffic coexist. While existing approaches employ dynamic routing and traffic prioritization, the integration of these techniques with intelligent control mechanisms remains underdeveloped, leading to suboptimal resource allocation and increased latency. Additionally, the complexity of hybrid switching architectures, which combine optical and electronic components, presents challenges in terms of reconfiguration delays, power consumption, and overall system reliability. These limitations are compounded by the lack of standardized frameworks that can seamlessly integrate optical and electronic switching, making it difficult to deploy and manage next-generation optical networks across different platforms and applications. To address these challenges, future research should focus on the development of intelligent and adaptive control frameworks that can dynamically optimize traffic routing, resource allocation, and network reconfiguration in real-time. This includes the integration of machine learning and artificial intelligence techniques to predict traffic patterns, anticipate network bottlenecks, and make proactive decisions to enhance overall performance. Additionally, the design of more efficient hybrid switching architectures that minimize latency and power consumption while maintaining scalability is a critical research direction. This can be achieved through the development of advanced optical switching technologies, such as reconfigurable optical add-drop multiplexers (ROADMs) and wavelength-selective

switches (WSS), which offer greater flexibility and faster reconfiguration times. Furthermore, the creation of standardized protocols and management frameworks that enable seamless interoperability between optical and electronic components will be essential for the widespread adoption of optical networking solutions in AI-driven data centers. The potential impact of the proposed future work extends beyond improving the performance and efficiency of optical networks. By enabling more intelligent and adaptive control mechanisms, the integration of AI and machine learning into optical networking can lead to significant advancements in network resilience, fault tolerance, and energy efficiency. This will not only enhance the scalability of data center infrastructures but also support the development of next-generation applications that require high-bandwidth, low-latency communication. Moreover, the adoption of advanced hybrid architectures and standardized frameworks will facilitate the deployment of optical networking solutions in a broader range of environments, from edge computing to cloud data centers. Ultimately, these advancements will contribute to the creation of more sustainable, agile, and intelligent optical networks that can meet the evolving demands of AI-driven applications and large-scale distributed systems.

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